

Rectangular Systems and Modules in Algebraic Number Fields. I

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Introduction

1. In what follows we fix an arbitrary finite algebraic number field $\mathfrak{K}$. Unless otherwise stated, only numbers and ideals of $\mathfrak{K}$ are considered, and they are integral numbers and integral ideals.

2. Let

$$A = (a_{ik}), \quad B = (b_{kl})$$

be two rectangular systems of integral numbers from $\mathfrak{K}$. If the number of columns of A is equal to the number of rows of B , their product is the rectangle $C = (c_{il})$ defined by

$$c_{il} = \sum_k a_{ik} b_{kl}.$$

Then C has as many rows as A and as many columns as B . This multiplication is associative, but not necessarily commutative. If two rectangles A, B satisfy a relation

$$B = PAQ,$$

then B is called divisible by A . If, in addition, A is divisible by B , then A and B are equivalent.

3. The first question treated here is the criterion for equivalence of two systems A, B ; in a second article the more general question of divisibility of B by A is to be treated. Over the rational field the answer is known. Over an arbitrary algebraic number field the final results remain the same for square systems of nonzero determinant, although the proof requires other tools. Once one allows nonsquare systems, or square systems of determinant 0, the results themselves change.

4. Left and right divisibility are special cases. B is left-divisible by A if $B = PA$, and right-divisible by A if $B = AQ$. Correspondingly, two systems are left- or right-equivalent if each is left- or right-divisible by the other.

Preliminaries

5. We recall the ideal-theoretic facts and notation that will be used. The nonzero ideals of $\mathfrak{K}$ are divided into finitely many classes; let their number be h . The class of the product of two ideals $\mathfrak{a}, \mathfrak{b}$ depends only on the classes of the factors:

$$\text{Kl}(\mathfrak{a}\mathfrak{b}) = \text{Kl}(\mathfrak{a}) \text{Kl}(\mathfrak{b}). \quad (1)$$

The h classes form an abelian group. The identity is the principal class, consisting of principal ideals. Fractional ideals are assigned to the same classes in such a way that (1) remains valid.

6. Any finite family of integral ideals $\mathfrak{a}_1, \dots, \mathfrak{a}_n$ has a greatest common divisor

$$(\mathfrak{a}_1, \dots, \mathfrak{a}_n) = \mathfrak{t},$$

which is a multiple of every common divisor. If all ideals are zero, then $\mathfrak{t} = 0$. For numbers $a_1, \dots, a_n$ this means the gcd of the principal ideals generated by them; the expressions

$$a_1x_1 + \dots + a_nx_n$$

represent precisely the numbers divisible by $(a_1, \dots, a_n)$. For integral or fractional ideals,

$$(\mathfrak{a}_1, \dots, \mathfrak{a}_n)\mathfrak{q} = (\mathfrak{a}_1\mathfrak{q}, \dots, \mathfrak{a}_n\mathfrak{q}) \quad (2)$$

and more generally

$$(\mathfrak{a}_1, \dots, \mathfrak{a}_m)(\mathfrak{b}_1, \dots, \mathfrak{b}_n) = (\mathfrak{a}_i\mathfrak{b}_j \mid 1 \leq i \leq m, 1 \leq j \leq n). \quad (3)$$

Even an infinite system of integral ideals has a greatest common divisor, already determined by some finite subsystem.

7. Let $\mathfrak{p}_1, \dots, \mathfrak{p}_l$ be distinct prime ideals and $e_1, \dots, e_l$ nonnegative rational integers. In every ideal class there are ideals that contain each $\mathfrak{p}_i$ exactly to the prescribed exponent e_i . In particular, in every class one can find ideals divisible by a given ideal, and also ideals relatively prime to a given ideal.

8. Let $a_1, \dots, a_n$ be not all zero and put

$$(a_1, \dots, a_n) = \mathfrak{t}.$$

As c ranges through the nonzero elements of $\mathfrak{K}$, the systems $a_1c, \dots, a_nc$ are precisely the systems proportional to $a_1, \dots, a_n$, and

$$(a_1c, \dots, a_nc) = \mathfrak{t}(c)$$

ranges over the ideals in the class $\text{Kl}(\mathfrak{t})$. Thus every system is proportional to an integral system whose elements have no nontrivial common divisor.

Elementary Divisors, Row Class and Column Class

9. If A is a rectangular system of integral numbers from $\mathfrak{K}$, the greatest common divisor $\mathfrak{d}_k$ of its k -rowed minors is called the k -th determinant divisor of A . If k is larger than the number of rows or columns, set $\mathfrak{d}_k = 0$. Laplace's determinant theorem implies that $\mathfrak{d}_k$ divides $\mathfrak{d}_{k+1}$. If $\mathfrak{d}_{r+1}$ is the first zero determinant divisor, r is the rank of A :

$$r = \text{Rg } A.$$

With $\mathfrak{d}_0 = (1)$, the ideals

$$\mathfrak{e}_k = \mathfrak{d}_k \mathfrak{d}_{k-1}^{-1}$$

are the elementary divisors of A .

10. If

$$B = PAQ,$$

then each determinant divisor of B is divisible by the corresponding determinant divisor of A . Consequently equivalent rectangular systems have the same determinant divisors, hence the same elementary divisors. This is the first condition for equivalence.

11. Let $A = (a_{ik})$ be an integral $m \times n$ rectangle of rank $r \geq 1$. Form a new rectangle A' from all r -minors of A , arranged so that its rows correspond to choices of r rows and its columns to choices of r columns. This rectangle has rank 1. The gcds of the nonzero rows of A' therefore belong to one ideal class; this is the row class

$$\text{Zkl } A.$$

The gcds of the columns similarly determine the column class

$$\text{Kkl } A.$$

12. If $\mathfrak{d}_r$ is the r -th determinant divisor of a rectangle A of rank r , then

$$\text{Zkl } A \cdot \text{Kkl } A = \text{Kl}(\mathfrak{d}_r). \quad (4)$$

Indeed, if one nonzero element of the rectangle of r -minors A' is written as uv , then every row of A' is proportional to the same system of v 's and every column to the same system of u 's. The common divisor of all elements of A' is exactly $\mathfrak{d}_r$.

13. If $B = PA$ and A, B have the same rank, then they have the same row class:

$$\text{Zkl } B = \text{Zkl } A.$$

Likewise, if $B = AQ$ and the ranks agree, the column class is preserved. This follows by expressing the r -minors of B as linear combinations of the r -minors of A by the general multiplication theorem for determinants.

14. If A and $B = PAQ$ are equivalent rectangular systems of rank $r \geq 1$, then they have the same determinant divisors, the same elementary divisors, and the same row and column classes:

$$\text{Zkl } A = \text{Zkl } B, \quad \text{Kkl } A = \text{Kkl } B. \quad (5)$$

This is the second condition for equivalence.

Linear Forms, Equations and Congruences

15. Consider m linear forms in n unknowns,

$$y_i = a_{i1}x_1 + \cdots + a_{in}x_n \quad (i = 1, \dots, m), \quad (6)$$

with integral coefficients in $\mathfrak{K}$. Let $\mathfrak{t}$ be the gcd of all coefficients a_{ik} . If all x_k are chosen divisible by an ideal $\mathfrak{u}$, then every y_i is divisible by $\mathfrak{tu}$. If the coefficient system $A = (a_{ik})$ has rank at least 2, the x_k can be chosen so that

$$(y_1, \dots, y_m) = \mathfrak{tu}. \quad (7)$$

The proof uses a finite set of prime ideals $\mathfrak{p}_1, \dots, \mathfrak{p}_l$ containing the prime factors of $\mathfrak{tu}$ and of one nonzero two-rowed determinant. First the x_k are chosen by simultaneous congruences

$$x_k \equiv \xi_{ik} \pmod{\mathfrak{p}_i^{e_i + f_i + 1}}$$

so that at each $\mathfrak{p}_i$ the gcd of the values has the prescribed exponent. Then a nonzero determinant

$$\Delta = a_{11}a_{22} - a_{12}a_{21}$$

is used to alter two variables without destroying those congruences, eliminating all remaining prime factors. This proves (7). For $\mathfrak{u} = (1)$, the values of the linear forms can therefore be made to have the same gcd as the coefficients.

16. Let A be a rectangle with n columns and $m \leq n - 2$ rows, of rank m , and let $\mathfrak{b}$ be its m -th determinant divisor. If one adjoins a new row $x_1, \dots, x_n$, with all entries divisible by a prescribed ideal $\mathfrak{u}$, the row can be chosen so that the $(m + 1)$ -st determinant divisor of the enlarged rectangle is

$$\mathfrak{bu}.$$

Without a divisibility condition on the new row, the determinant divisor can be kept equal to $\mathfrak{b}$. Iterating gives: a rectangle with n columns and $m < n$ rows whose m -minors are coprime can be completed by further rows to a square system of determinant 1.

17. If the coefficient system in (6) has rank at most $n - 2$, the equations

$$y_i = 0 \quad (i = 1, \dots, m)$$

have a solution in coprime numbers. Choose two linearly independent integral solutions $\alpha = (\alpha_k)$ and $\beta = (\beta_k)$, arrange that their gcds are relatively prime, and then choose λ, μ so that the numbers

$$\alpha_k \lambda + \beta_k \mu$$

are coprime.

18. If the coefficient system in (6) has rank at most $n - 1$, and if $\mathfrak{m} \neq 0$ is an ideal, the congruences

$$y_i \equiv 0 \pmod{\mathfrak{m}} \quad (i = 1, \dots, m) \quad (8)$$

can be solved in coprime numbers. One first chooses a nontrivial integral solution of $y_i = 0$ whose gcd is relatively prime to $\mathfrak{m}$, and then replaces its entries by congruent integral numbers modulo $\mathfrak{m}$ so as to obtain an actually coprime solution of the congruences.

19. Let $\mathfrak{e}_n$ be the n -th elementary divisor of the coefficient system $A = (a_{ik})$ in (6). Every solution of the congruences (8) consists of numbers divisible by

$$\frac{\mathfrak{m}}{(\mathfrak{m}, \mathfrak{e}_n)}.$$

If $\mathfrak{e}_n = 0$, this is trivial. Otherwise let $\mathfrak{b}_n$ and $\mathfrak{b}_{n-1}$ be the n -th and $(n - 1)$ -st determinant divisors, so that

$$\mathfrak{e}_n = \mathfrak{b}_n \mathfrak{b}_{n-1}^{-1}.$$

From any nonsingular subsystem of n equations, Cramer's rule expresses each x_k as a quotient whose denominator is an n -rowed determinant and whose numerator is divisible by $\mathfrak{b}_{n-1} \mathfrak{m}$. Letting the nonzero n -determinants range over all such subsystems and using their gcd $\mathfrak{b}_n$ yields the asserted divisibility.

20. The homogeneous linear congruences

$$\sum_{k=1}^n a_{ik} x_k \equiv 0 \pmod{\mathfrak{m}} \quad (i = 1, \dots, m) \quad (9)$$

are solvable in coprime numbers if and only if $\mathfrak{m}$ divides the n -th elementary divisor $\mathfrak{e}_n$ of the coefficient system. Necessity is Nr. 19. For sufficiency one reduces to $\mathfrak{m} = \mathfrak{e}_n$, factors $\mathfrak{e}_n$ into prime-ideal powers, solves the congruences for each prime-ideal power by Nr. 18, and combines the resulting solutions by simultaneous congruences. The final numbers may then be replaced by congruent integral numbers so that their gcd is 1.

21. The system of equations

$$\sum_{k=1}^n a_{ik} x_k = c_i \quad (i = 1, \dots, m) \quad (22)$$

is solvable in integral numbers if and only if the coefficient system $A = (a_{ik})$ and the system A' , obtained from A by adjoining $c_1, \dots, c_m$ as last column, have the same rank r and the same r -th determinant divisor $\mathfrak{b}$.

Proof. If $\xi_1, \dots, \xi_n$ are an integral solution of (22), then elementary determinant theory shows that every r -minor of A' which contains the last column is a linear form in the r -minors of A as coefficients; hence it is divisible by $\mathfrak{b}$. Thus, in passing from A to A' , the r -th determinant divisor, and likewise every other determinant divisor, remains unchanged. Similarly, the rank is not increased. The stated conditions are therefore necessary.

Conversely assume that the conditions are fulfilled, and let δ be a nonzero r -minor of A . Order the equations and unknowns so that

$$\begin{vmatrix} a_{11} & \cdots & a_{1r} \\ \vdots & & \vdots \\ a_{r1} & \cdots & a_{rr} \end{vmatrix} = \delta.$$

Then the equations

$$\sum_{k=1}^r a_{ik}x_k = \delta c_i \quad (i = 1, \dots, r)$$

have an integral solution $\xi_1, \dots, \xi_r$. Since r is the rank of the whole coefficient system, the equations

$$\sum_{k=1}^r a_{ik}\xi_k = \delta c_i$$

hold; or, putting $\xi_{r+1} = \xi_{r+2} = \cdots = \xi_n = 0$, we have

$$\sum_{k=1}^n a_{ik}\xi_k = \delta c_i \quad (i = 1, \dots, m). \quad (23)$$

Let $\mathfrak{p}_1, \mathfrak{p}_2, \dots, \mathfrak{p}_l$ be the prime factors occurring in δ , and fix one of them, $\mathfrak{p}_q$. Choose an r -minor δ_q of A in which $\mathfrak{p}_q$ occurs as often as in δ . Let $\delta_q = r_q \delta$ be the principal ideal represented by δ_q ; then r_q is not divisible by $\mathfrak{p}_q$. Finally choose a number μ_q divisible by r_q but not by $\mathfrak{p}_q$. Again ordering equations and unknowns suitably, suppose that

$$\begin{vmatrix} a_{11} & \cdots & a_{1r} \\ \vdots & & \vdots \\ a_{r1} & \cdots & a_{rr} \end{vmatrix} = \delta_q.$$

Then the solutions of

$$\sum_{k=1}^r a_{ik}x_k = \mu_q c_i \quad (i = 1, \dots, r)$$

are integral; setting all other unknowns equal to 0, we obtain equations of the form

$$\sum_{k=1}^n a_{ik}\xi_{qk} = \mu_q c_i \quad (i = 1, \dots, m). \quad (24)$$

This can be done for every prime factor $\mathfrak{p}_1, \dots, \mathfrak{p}_l$. Then $\delta, \mu_1, \dots, \mu_l$ are coprime. Choose $\alpha, \beta_1, \dots, \beta_l$ so that

$$\alpha\delta + \beta_1\mu_1 + \cdots + \beta_l\mu_l = 1, \quad (25)$$

and set

$$\alpha\xi_k + \beta_1\xi_{1k} + \cdots + \beta_l\xi_{lk} = \bar{\xi}_k \quad (k = 1, \dots, n). \quad (26)$$

Equations (23), (24), (25), and (26) give

$$\sum_{k=1}^n a_{ik}\bar{\xi}_k = c_i \quad (i = 1, \dots, m), \quad (27)$$

which proves the theorem.¹

¹This last argument assumes $n > 1$. For $n = 1$ the assertion follows immediately, since then $x = 1$ satisfies the conditions of the problem.

§ 4. Modules

22. From now on the degree of a rectangle means its number of columns. This applies in particular when the rectangle consists of only one row; its degree is then the number of its elements. We use small Greek letters for rows. The equation $\sigma = 0$ means that all elements of σ are zero. A factor of a row σ is any ideal which divides every element of σ ; the greatest common divisor of the elements is called the greatest factor of σ . By $a\sigma$ we mean the row obtained from σ by multiplying its entries by the number a . If σ and τ have the same degree, then $\sigma + \tau$ and $\sigma - \tau$ denote the rows obtained by componentwise addition and subtraction. If integral rows $\sigma, \sigma_1, \sigma_2, \dots, \sigma_m$ satisfy a relation

$$\sigma = a_1\sigma_1 + \dots + a_m\sigma_m$$

with integral coefficients a_i , then σ is said to be composable from $\sigma_1, \dots, \sigma_m$.

23. Modules. A system $\mathfrak{S}$ of integral rows of degree n is called a module of degree n if it has the following two properties:

- 1) if $\sigma \in \mathfrak{S}$, then $a\sigma \in \mathfrak{S}$ for every integral number a of $\mathfrak{K}$;
- 2) if $\sigma, \tau \in \mathfrak{S}$, then $\sigma + \tau \in \mathfrak{S}$.

Modules are denoted by capital German letters; a lower index gives the degree. The row $\sigma = 0$ by itself forms a module, the zero module.

Two rows σ, τ are congruent modulo a module $\mathfrak{A}_n$, written

$$\sigma \equiv \tau \pmod{\mathfrak{A}_n},$$

if $\sigma - \tau$ belongs to the module. The module $\mathfrak{A}_n$ is said to be divisible by the module $\mathfrak{B}_n$ if every row of $\mathfrak{A}_n$ is also contained in $\mathfrak{B}_n$; then $\mathfrak{B}_n$ is a divisor of $\mathfrak{A}_n$, and $\mathfrak{A}_n$ is a multiple of $\mathfrak{B}_n$. A congruence which holds modulo $\mathfrak{A}_n$ also holds modulo every divisor of $\mathfrak{A}_n$.

24. Rank, determinant divisors, and elementary divisors of a module. Let a module $\mathfrak{A}_n$ be given. Consider all rectangles A formed from rows of $\mathfrak{A}_n$. If among them there are rectangles of rank r , but none of rank $r + 1$, then r is called the rank of the module, $\text{Rg } \mathfrak{A}_n$. For each natural number k , the k -th determinant divisors of all such rectangles A have a greatest common divisor $\mathfrak{b}_k$; this is the k -th determinant divisor of $\mathfrak{A}_n$. Clearly $\mathfrak{b}_k$ divides $\mathfrak{b}_{k+1}$, and if $\text{Rg } \mathfrak{A}_n = r$, then the first r determinant divisors are nonzero and all following ones are zero. The elementary divisors of a module are then defined as before for rectangular systems.

25. Basis of a module. If $\sigma_1, \sigma_2, \dots, \sigma_m$ are rows of degree n , the collection of all rows composable from them is plainly a module of degree n . Conversely, every module $\mathfrak{A}_n$ is generated by finitely many of its rows. Indeed, let $\text{Rg } \mathfrak{A}_n = r$, and let $\mathfrak{b}_r$ be its r -th determinant divisor. From the rectangles made of rows of $\mathfrak{A}_n$ choose an r -rowed rectangle A_0 whose r -th determinant divisor $\mathfrak{b}_{0r}$ is nonzero. Let $\mathfrak{p}_1, \dots, \mathfrak{p}_l$ be the distinct prime factors of $\mathfrak{b}_{0r}$. By the definition of determinant divisors for modules, we can choose rectangles $A_1, \dots, A_l$, each made of r rows from $\mathfrak{A}_n$, so that the r -th determinant divisor $\mathfrak{b}_{qr}$ of A_q contains $\mathfrak{p}_q$ exactly as often as it occurs in $\mathfrak{b}_r$. Then

$$(\mathfrak{b}_{0r}, \mathfrak{b}_{1r}, \dots, \mathfrak{b}_{lr}) = \mathfrak{b}_r,$$

and the rectangle made from the $(l + 1)r$ rows of $A_0, A_1, \dots, A_l$ has $\mathfrak{b}_r$ as its r -th determinant divisor.

Now let

$$A = (a_{ik}) \quad (i = 1, \dots, m; \ k = 1, \dots, n)$$

be any rectangle whose rows, also denoted $\sigma_1, \dots, \sigma_m$, belong to $\mathfrak{A}_n$, and whose r -th determinant divisor is $\mathfrak{b}_r$. If A is enlarged by adjoining any row $\sigma = (c_1, \dots, c_n)$ from $\mathfrak{A}_n$, neither the rank nor the r -th determinant divisor changes. Hence, by No. 21, the equations

$$\sum_{i=1}^m a_{ik} a_i = c_k \quad (k = 1, \dots, n)$$

are solvable in integral numbers $a_1, \dots, a_m$, and therefore

$$\sigma = a_1 \sigma_1 + \dots + a_m \sigma_m.$$

Thus every row of the module is composable from $\sigma_1, \dots, \sigma_m$.

A system of rows $\sigma_1, \dots, \sigma_m$ of a module $\mathfrak{A}_n$, from which all rows of the module can be composed, is called a basis of $\mathfrak{A}_n$; the same name is used for the rectangle A formed from these rows. We write

$$\mathfrak{A}_n = \text{Md}(\sigma_1, \dots, \sigma_m) = \text{Md } A.$$

The basis is irreducible if there is no basis with fewer rows.

26. Let $\mathfrak{A}_n = \text{Md } A$ and $\mathfrak{B}_n = \text{Md } B$ be two modules of degree n . For $\mathfrak{B}_n$ to be divisible by $\mathfrak{A}_n$, it is necessary and sufficient that the rows of B can be composed from those of A , or equivalently that

$$B = PA$$

for some integral matrix P ; that is, B is left-divisible by A . Consequently,

$$\text{Md } B = \text{Md } A$$

holds if and only if A and B are left-equivalent.

If C is a rectangle made of k rows of $\mathfrak{A}_n$, then $C = PA$, and the k -th determinant divisor of A divides that of C . From this and from the definition of determinant divisors for modules it follows that a basis A has the same determinant divisors, hence the same elementary divisors, as the module $\mathfrak{A}_n$.

All bases of a module are equivalent; hence they have the same row class and the same column class. We therefore transfer these notions to the module. Of the two classes we shall prefer the latter, and call it simply the class of the module:

$$\text{Kkl } \mathfrak{A}_n = \text{Kl } A.$$

Thus the modules fall into as many classes as the ideals of the field $\mathfrak{K}$.

27. Let $A = (a_{ik})$ be a rectangle made of m rows $\sigma_1, \dots, \sigma_m$ of degree n , let $\text{Rg } A = r$, $\text{Md } A = \mathfrak{A}$, and let $a_1, \dots, a_m$ be coprime numbers. Put

$$a_1 \sigma_1 + \dots + a_m \sigma_m = \sigma. \tag{28}$$

Then there is a square system P of degree m , determinant 1, whose first row is $a_1, \dots, a_m$. The inverse system P^{-1} is also integral; thus $B = PA$ and $A = P^{-1}B$ are left-equivalent. Consequently $B = PA$, like A , is an m -rowed basis of $\mathfrak{A}$, and σ is the first row of this basis. If $a_1, \dots, a_m$ can be chosen so that $\sigma = 0$, then deleting this zero row gives an $(m-1)$ -rowed basis of $\mathfrak{A}$; the basis $\sigma_1, \dots, \sigma_m$ is then reducible.

No. 17 says that, when $r \leq m-2$, the equations

$$\sum_{i=1}^m a_{ki} x_i = 0 \quad (k = 1, \dots, n)$$

or equivalently

$$x_1\sigma_1 + \cdots + x_m\sigma_m = 0$$

can be satisfied by coprime numbers. Hence an irreducible basis of a module of rank r cannot contain more than $r + 1$ rows. It plainly cannot contain fewer than r rows, so the remaining question is when it has r rows and when it has $r + 1$.

First consider a nonzero module $\mathfrak{A}$ of rank r with an r -rowed basis A . Form, as in No. 11, the rectangle A' of r -minors of A . This rectangle consists of a single row; each column has one element. Thus the gcd of the elements of a column is a principal ideal, and the column class of $\mathfrak{A}$ is the principal class. Conversely, suppose that a rank r module $\mathfrak{A}$ belongs to the principal class, and let $A = (a_{ik})$ be a basis of $\mathfrak{A}$ consisting of rows $\sigma_1, \dots, \sigma_r, \sigma_{r+1}$. If $c_1, \dots, c_{r+1}$ is a nonzero column of the rectangle A' of r -minors of A , then

$$c_1\sigma_1 + \cdots + c_{r+1}\sigma_{r+1} = 0 \quad (29)$$

and

$$(c_1, c_2, \dots, c_{r+1}) \quad (30)$$

is a principal ideal, represented by an integral number a . The numbers $c'_k = c_k/a$ are integral and coprime, and

$$c'_1\sigma_1 + \cdots + c'_{r+1}\sigma_{r+1} = 0.$$

Therefore the basis $\sigma_1, \dots, \sigma_{r+1}$ is reducible, and $\mathfrak{A}$ has an r -rowed basis.

Thus: an irreducible basis of a rank r module $\mathfrak{A}$ consists of r rows when $\mathfrak{A}$ belongs to the principal class, and otherwise of $r + 1$ rows.

§ 5. The Equivalence Conditions

28. In § 2 we gave two necessary conditions for the equivalence of rectangular systems. We now show that together they are also sufficient. The proof rests on a normal form for a module basis. This normal form is not uniquely fixed, but it is characterized by simple properties.

We first record the required consequences of the preceding work in a slightly altered form. Let

$$A = (a_{ik}) \quad (i = 1, \dots, m; k = 1, \dots, n)$$

be a rectangular system with rows $\sigma_1, \dots, \sigma_m$ and n columns. Let $\text{Md } A = \mathfrak{A}$, $\text{Rg } A = r \geq 1$, let $\mathfrak{b}_r$ be the r -th determinant divisor and $\mathfrak{e}_r$ the r -th elementary divisor of A . Let σ' denote any row of the form

$$\sigma' = c_1\sigma_1 + \cdots + c_m\sigma_m,$$

where the coefficients satisfy

$$(c_1, \dots, c_m) = 1 \quad (31)$$

and are otherwise arbitrary. Then:

- a) $\mathfrak{A}$ has a basis consisting of any prescribed row of the form σ' and $m - 1$ further rows;
- b) if $m \geq r + 1$ and $\mathfrak{m} \neq 0$, there are rows σ' having $\mathfrak{m}$ as a factor;
- c) if $m = r$, the greatest factor of a row σ' divides $\mathfrak{e}_r$, and there are rows σ' for which it equals $\mathfrak{e}_r$;
- d) a row σ of degree n belongs to $\mathfrak{A}$ exactly when adjoining it to A gives a rectangle of rank r and r -th determinant divisor $\mathfrak{b}_r$;
- e) a rectangle B made of rows of $\mathfrak{A}$ is a basis of $\mathfrak{A}$ exactly when its r -th determinant divisor is $\mathfrak{b}_r$.

29. Modules of the principal class. Let $\mathfrak{A} = \mathfrak{M}^{(r)}$ be a module of rank $r \geq 1$ belonging to the principal class. It has a basis

$$\sigma_1, \sigma_2, \dots, \sigma_r \quad (32)$$

of r rows. Let $\mathfrak{e}_r$ be the r -th elementary divisor of $\mathfrak{M}^{(r)}$. From a) and c) of the preceding number, the basis may be chosen so that the greatest factor of σ_r is $\mathfrak{e}_r$. Assume this and set

$$\text{Md}(\sigma_1, \dots, \sigma_{r-1}) = \mathfrak{M}^{(r-1)}.$$

Replacing the first $r - 1$ rows of (32) by any basis of $\mathfrak{M}^{(r-1)}$ again gives a basis of $\mathfrak{A}$. If $\mathfrak{e}_{r-1}$ is the $(r - 1)$ -st elementary divisor of $\mathfrak{M}^{(r-1)}$, the same argument shows that $\mathfrak{M}^{(r-1)}$ has a basis of $r - 1$ rows whose last row has greatest factor $\mathfrak{e}_{r-1}$. Continuing, we obtain a basis $\sigma_1, \dots, \sigma_r$ such that the greatest factor $\mathfrak{e}_k$ of σ_k is the k -th elementary divisor of

$$\mathfrak{M}^{(k)} = \text{Md}(\sigma_1, \dots, \sigma_k).$$

Assume such a basis. If $c_1, \dots, c_k$ are coprime, then by No. 28 c) the greatest factor of

$$\sigma' = c_1\sigma_1 + \dots + c_k\sigma_k \quad (33)$$

divides $\mathfrak{e}_k$. Taking $k > 1$, $c_{k-1} = 1$, and all other coefficients zero gives $\sigma' = \sigma_{k-1}$, whose greatest factor is $\mathfrak{e}_{k-1}$. Thus $\mathfrak{e}_{k-1}$ divides $\mathfrak{e}_k$.

Let A_k be the rectangle made from $\sigma_1, \dots, \sigma_k$. Let $\mathfrak{b}_k$ be its k -th determinant divisor and, for $k > 1$, let $\mathfrak{b}'_{k-1}$ be its $(k - 1)$ -st determinant divisor. Then

$$\mathfrak{b}_1 = \mathfrak{e}_1, \quad \mathfrak{b}_k = \mathfrak{b}'_{k-1}\mathfrak{e}_k. \quad (34)$$

Since the last row of A_k has factor $\mathfrak{e}_k$, and the $(k - 1)$ -minors made from the first $k - 1$ rows have factor $\mathfrak{b}_{k-1}$, it follows that $\mathfrak{b}_k$ is divisible by $\mathfrak{b}_{k-1}\mathfrak{e}_k$. Together with (34), this gives $\mathfrak{b}'_{k-1} = \mathfrak{b}_{k-1}$, and hence

$$\mathfrak{b}_k = \mathfrak{e}_1\mathfrak{e}_2 \cdots \mathfrak{e}_k \quad (k = 1, \dots, r). \quad (35)$$

Because $\mathfrak{e}_{k-1}$ divides $\mathfrak{e}_k$, every product of k factors chosen from $\mathfrak{e}_1, \dots, \mathfrak{e}_r$ is divisible by $\mathfrak{b}_k$; therefore every k -minor of $A = A_r$ has factor $\mathfrak{b}_k$. Conversely, $\mathfrak{b}_k$ already appears as the gcd of the k -minors of A_k . Hence $\mathfrak{b}_k$ and $\mathfrak{e}_k$ are the k -th determinant and elementary divisors of A and of $\mathfrak{A}$.

Thus a principal-class module of rank r has a basis $\sigma_1, \dots, \sigma_r$ whose greatest factors are precisely the elementary divisors $\mathfrak{e}_1, \dots, \mathfrak{e}_r$ of the module; each of these divides the next.

30. Normal basis of an arbitrary module. Let $\mathfrak{A}$ be any module of rank $r \geq 1$, with determinant divisors $\mathfrak{b}_k$ and elementary divisors $\mathfrak{e}_k$. It has a basis of $r + 1$ rows $\sigma_1, \dots, \sigma_r, \sigma_{r+1}$. By No. 28 a), b), this basis may be chosen so that the entries of σ_{r+1} are divisible by any prescribed nonzero ideal $\mathfrak{m}$. We take σ_{r+1} to have the factor $2\mathfrak{b}_r^2$.

Let A_k be the rectangle made from $\sigma_1, \dots, \sigma_k$, and let $\mathfrak{b}'_k$ be the k -th determinant divisor of A_k . Then $\mathfrak{b}'_k$ is divisible by $\mathfrak{b}_k$, and $\mathfrak{b}_k$ is the gcd of $\mathfrak{b}'_k$ and the k -minors of A_{k+1} in which entries of σ_{r+1} occur. The latter are all divisible by $2\mathfrak{b}_r^2$; hence $\mathfrak{b}'_k$ must contain each prime factor of $2\mathfrak{b}_k$ exactly as often as $\mathfrak{b}_k$. Therefore

$$\mathfrak{b}'_k = \mathfrak{b}_k\mathfrak{q}_k$$

with $\mathfrak{q}_k$ relatively prime to $2\mathfrak{b}_k$. For $k < r$, inspection of the $(k + 1)$ -minors gives $\mathfrak{q}_k = 1$. Thus A_r has determinant divisors

$$\mathfrak{b}_1, \dots, \mathfrak{b}_{r-1}, \mathfrak{b}_r\mathfrak{q}_r$$

and elementary divisors

$$\mathfrak{e}_1, \dots, \mathfrak{e}_{r-1}, \mathfrak{e}_r\mathfrak{q}_r. \quad (36)$$

Since $\text{Md } A_r$ is a principal-class module, No. 29 implies that each ideal in (36) divides the next; because $\mathfrak{q}_r$ is prime to $\mathfrak{e}_{r-1}$, it follows that $\mathfrak{e}_{r-1}$ divides $\mathfrak{e}_r$. Thus, in general, the elementary divisors

$$\mathfrak{e}_1, \mathfrak{e}_2, \dots$$

of a rectangle or module form a system of ideals in which each divides the next.

31. Continuation. We now transform the basis $\sigma_1, \dots, \sigma_r, \sigma_{r+1}$ of the preceding module. First replace $\sigma_1, \dots, \sigma_r$ by another r -rowed basis of $\text{Md}(\sigma_1, \dots, \sigma_r)$. This module belongs to the principal class and has elementary divisors

$$\mathfrak{e}_1, \dots, \mathfrak{e}_{r-1}, \mathfrak{e}_r \mathfrak{q}$$

where $\mathfrak{q} = \mathfrak{q}_r$. Hence, by No. 29, we may choose the new rows so that their greatest factors are precisely these elementary divisors. We again denote the rows by $\sigma_1, \dots, \sigma_r$, the rectangle they form by A_r , and the rectangle obtained by adjoining σ_{r+1} by A_{r+1} , so that $\mathfrak{A} = \text{Md } A_{r+1}$.

Let $\mathfrak{q}'$ be an ideal of the same class as $\mathfrak{q}$, relatively prime to $\mathfrak{q}$. Then there is a row σ'_r , proportional to σ_r , with greatest factor $\mathfrak{e}_r \mathfrak{q}'$. Adjoining this row to A_{r+1} gives a rectangle A_{r+2} of the same rank r . Its $r+2$ rows have factors

$$\mathfrak{e}_1, \dots, \mathfrak{e}_{r-1}, \mathfrak{e}_r \mathfrak{q}, \mathfrak{b}_r^2, \mathfrak{e}_r \mathfrak{q}',$$

and every product of r of these factors is divisible by $\mathfrak{e}_1 \cdots \mathfrak{e}_r = \mathfrak{b}_r$. Therefore A_{r+2} , like A_{r+1} , has r -th determinant divisor $\mathfrak{b}_r$, and by No. 28 d), σ'_r belongs to $\mathfrak{A}$.

Now consider the rectangle A'_{r+1} whose rows are

$$\sigma_1, \sigma_2, \dots, \sigma_r, \sigma'_r.$$

Its r -minors either vanish, or have gcd $\mathfrak{b}_r \mathfrak{q}$, or have gcd $\mathfrak{b}_r \mathfrak{q}'$. Since $(\mathfrak{q}, \mathfrak{q}') = 1$, the r -th determinant divisor is $\mathfrak{b}_r$. By No. 28 e), these rows form a basis of $\mathfrak{A}$. Hence every module $\mathfrak{A}$ of rank r with elementary divisors $\mathfrak{e}_1, \dots, \mathfrak{e}_r$ has a basis of $r+1$ rows

$$\sigma_1, \sigma_2, \dots, \sigma_r, \sigma'_r$$

such that $\sigma_1, \dots, \sigma_{r-1}$ have greatest factors $\mathfrak{e}_1, \dots, \mathfrak{e}_{r-1}$, while σ_r, σ'_r are proportional and the gcd of their $2n$ entries is $\mathfrak{e}_r$. Such a basis will be called normal.

32. Let $\mathfrak{A}$ be a module of rank $r \geq 1$, and let A be a basis of $\mathfrak{A}$ consisting of rows $\sigma_1, \dots, \sigma_r, \sigma_{r+1}$. There exist $r+1$ numbers $c_1, \dots, c_{r+1}$, not all zero, such that

$$c_1 \sigma_1 + \cdots + c_{r+1} \sigma_{r+1} = 0. \quad (37)$$

These numbers are determined up to a common factor. If one forms the rectangle of r -minors of A , its nonzero columns are proportional to $c_1, \dots, c_{r+1}$. Thus the class $\mathfrak{x}$ of the gcd

$$\mathfrak{t} = (c_1, \dots, c_{r+1})$$

is also the column class of $\mathfrak{A}$.

If A is normal, the last two rows, again denoted σ_r, σ'_r , are proportional, and (37) takes the simple form

$$c_r \sigma_r + c'_r \sigma'_r = 0, \quad (38)$$

so that

$$\text{Kl } \mathfrak{A} = \text{Kl}(c_r, c'_r). \quad (39)$$

Let $\mathfrak{e}_r$ be the r -th determinant divisor of $\mathfrak{A}$, the gcd of the $2n$ entries of σ_r, σ'_r . If τ, τ' are two rows proportional to σ_r and having the same gcd $\mathfrak{e}_r$ among their $2n$ entries, then

$$\text{Md}(\sigma_r, \sigma'_r) = \text{Md}(\tau, \tau'). \quad (40)$$

Hence replacing σ_r, σ'_r by τ, τ' again gives a normal basis of $\mathfrak{A}$.

Let $\mathfrak{e}_r \mathfrak{q}$ and $\mathfrak{e}_r \mathfrak{q}'$ be the greatest factors of σ_r and σ'_r . Then

$$(\mathfrak{q}, \mathfrak{q}') = 1, \quad (41)$$

and $\mathfrak{q}, \mathfrak{q}'$ lie in the same class λ . If c_r, c'_r denote also the principal ideals represented by the corresponding numbers, then (38) gives

$$\mathfrak{q} c_r = \mathfrak{q}' c'_r.$$

Together with (41), this yields

$$c_r = c_r(\mathfrak{q}, \mathfrak{q}') = (\mathfrak{q} c_r, \mathfrak{q}' c_r) = (\mathfrak{q}' c'_r, \mathfrak{q}' c_r) = \mathfrak{q}'(c_r, c'_r),$$

so that $\text{Kl } \mathfrak{q} = \lambda = 1/\varkappa$, or

$$\text{Kl } \mathfrak{A} = \text{Kl } \frac{1}{\mathfrak{q}}. \quad (42)$$

Now let c, c' be any two numbers whose gcd $\mathfrak{u}$ belongs to the class $\varkappa$, and let c, c' also denote the principal ideals they represent. Put

$$\mathfrak{t} = \frac{c'}{\mathfrak{u}}, \quad \mathfrak{t}' = \frac{c}{\mathfrak{u}}.$$

Then $\mathfrak{t}, \mathfrak{t}'$ are integral ideals in the class $1/\varkappa$, and $(\mathfrak{t}, \mathfrak{t}') = 1$. Hence one can choose two rows τ, τ' , proportional to σ_r , whose greatest factors are $\mathfrak{e}_r \mathfrak{t}$ and $\mathfrak{e}_r \mathfrak{t}'$. The rows $c\tau$ and $c'\tau'$ then have the same greatest factor $\mathfrak{e}_r c c' / \mathfrak{u}$; they therefore differ only by a unit, and may be chosen so that

$$c\tau + c'\tau' = 0.$$

Thus: if $\mathfrak{A}$ is a module of rank r , $\text{Kl } \mathfrak{A} = \varkappa$, and c, c' are any two numbers whose gcd belongs to $\varkappa$, then a normal basis

$$\sigma_1, \dots, \sigma_r, \sigma'_r$$

of $\mathfrak{A}$ can be chosen so that

$$c\sigma_r + c'\sigma'_r = 0.$$

33. The equivalence theorem. The preceding results give the theorem: two rectangular systems A and B are equivalent if and only if they have the same elementary divisors, the same row class, and the same column class.²

Proof. The necessity of these conditions was already proved in Nos. 10 and 14. Moreover the conditions are not independent: by No. 12, if two rectangles have the same elementary divisors and column class, then they must also have the same row class, and conversely with rows and columns interchanged.

Let A and B be two rectangles, of equal or unequal degree, of rank $r \geq 1$, with elementary divisors $\mathfrak{e}_1, \mathfrak{e}_2, \dots$, determinant divisors $\mathfrak{b}_1, \mathfrak{b}_2, \dots$, and common column class $\varkappa$. Choose nonzero numbers c, c' whose gcd belongs to the class $\varkappa$. Choose a normal basis

$$\sigma_1, \dots, \sigma_r, \sigma'_r$$

of $\mathfrak{A} = \text{Md } A$, and a normal basis

$$\tau_1, \dots, \tau_r, \tau'_r$$

²Rectangles of rank 0 belong to no row or column class. It is immediate that all such rectangles, whose entries, elementary divisors, and determinant divisors all vanish, are mutually equivalent. They may therefore be disregarded below.

of $\mathfrak{B} = \text{Md } B$, so that

$$c\sigma_r + c'\sigma'_r = 0, \quad c\tau_r + c'\tau'_r = 0. \quad (42)$$

Let

$$\begin{aligned} A' &= (a_{ik}) \quad (i = 1, \dots, r+1; k = 1, \dots, n), \\ B' &= (b_{ik}) \quad (i = 1, \dots, r+1; k = 1, \dots, n') \end{aligned}$$

be the rectangles formed from these rows. Then A' is left-equivalent to A , and B' to B . It remains only to prove that A' and B' are equivalent.

Adjoin to A' a column $b_{1q}, \dots, b_{r+1,q}$ of B' , forming A'_q . By (42), the last two rows of A'_q are still proportional, so $\text{Rg } A'_q = r$. Its rows have factors

$$\mathfrak{e}_1, \mathfrak{e}_2, \dots, \mathfrak{e}_r, \mathfrak{e}_r,$$

and all r -minors of A'_q are divisible by $\mathfrak{b}_r = \mathfrak{e}_1 \cdots \mathfrak{e}_r$. Hence the r -th determinant divisor is unchanged in passing from A' to A'_q . By No. 21 the equations

$$\sum_{k=1}^n a_{ik}x_k = b_{iq} \quad (i = 1, \dots, r+1)$$

are solvable in integral numbers $p_{1q}, \dots, p_{nq}$. Doing this for all n' columns of B' gives

$$\sum_{k=1}^n a_{ik}p_{kq} = b_{iq} \quad (i = 1, \dots, r+1; q = 1, \dots, n'). \quad (43)$$

With $P = (p_{ik})$, this is

$$A'P = B'.$$

The same argument proves the existence of $B'Q = A'$. Thus A' and B' are right-equivalent, and the theorem is proved.

34. The elementary divisors

$$\mathfrak{e}_1, \mathfrak{e}_2, \dots$$

of a rectangle or module form, as shown above, a sequence of ideals each dividing the next, with only finitely many nonzero terms. Conversely every such sequence is the elementary-divisor system of modules and rectangles, indeed of modules of every class; it will therefore be called simply an elementary-divisor system.

Let $\mathfrak{e}_1, \dots, \mathfrak{e}_r$ be nonzero ideals, each dividing the next, and let $\mathfrak{x}$ be an arbitrary ideal class. Let $\mathfrak{q}, \mathfrak{q}'$ be two ideals in the class reciprocal to $\mathfrak{x}$, with $(\mathfrak{q}, \mathfrak{q}') = 1$, and put

$$\mathfrak{e}_1 \mathfrak{e}_2 \cdots \mathfrak{e}_k = \mathfrak{b}_k \quad (k = 1, \dots, r).$$

If $n > r$, then by No. 16 we can construct a rectangle A_r of degree n , with r rows $\sigma_1, \dots, \sigma_r$, such that these rows have factors

$$\mathfrak{e}_1, \dots, \mathfrak{e}_{r-1}, \mathfrak{e}_r \mathfrak{q} \quad (44)$$

and the rectangle A_k made from the first k rows, for $k < r$, has k -th determinant divisor $\mathfrak{b}_k$, while the r -th determinant divisor of A_r is $\mathfrak{b}_r \mathfrak{q}$. It is then immediate that the ideals in (44) are the elementary divisors of A_r and also the greatest factors of its rows. Since $\mathfrak{q}, \mathfrak{q}'$ lie in the same class, there is a row σ'_r , proportional to σ_r , whose greatest factor is $\mathfrak{e}_r \mathfrak{q}'$. The rectangle A_{r+1} made from $\sigma_1, \dots, \sigma_r, \sigma'_r$ then has elementary divisors

$$\mathfrak{e}_1, \dots, \mathfrak{e}_r,$$

and so does the module $\mathfrak{A} = \text{Md } A_{r+1}$. Since $\mathfrak{q}, \mathfrak{q}'$ are in the class $1/\varkappa$, No. 32 gives

$$\text{Kl } \mathfrak{A} = \varkappa.$$

If one wants a module of degree and rank r with elementary divisors $\mathfrak{e}_1, \dots, \mathfrak{e}_r$, then its class cannot be prescribed arbitrarily: the row class of such a module is plainly the principal class, so its column class must be $\text{Kl } \mathfrak{b}_r$. Modules of this class can also be constructed. By No. 16, form a rectangle A' of r rows and degree $r+1$ whose rows have factors $\mathfrak{e}_1, \dots, \mathfrak{e}_r$ and whose r -th determinant divisor is $\mathfrak{b}_r$. Then A' has elementary divisors $\mathfrak{e}_1, \dots, \mathfrak{e}_r$; the same is true of the transposed rectangle A and of the degree r module $\mathfrak{A} = \text{Md } A$.